

SOUTH ALEUTIANS BUOY 46003, USA

WMO: 992170

Lat: 51.831N

Lon: 155.850W

Elev: 3

StdP: 101.29

Time Zone: -10.00 (W10)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.7	-1.3	N/A	N/A	N/A	N/A	N/A	N/A	17.0	1.9	16.4	2.2	8.7	300	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 1.1	(c) 13.2	(d) N/A	(e) 12.8	(f) N/A	(g) 12.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 7.9	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	15.2	14.2	DB	-4.1	14.1	1.1	1.7	-4.9	15.4	-5.6	16.3	-6.2	17.3	-7.0	18.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	3.2	2.7	2.9	3.2	5.1	7.1	9.5	11.3	10.8	8.1	5.5	4.1
	DBStd	3.34	1.63	2.04	1.38	1.08	0.82	1.00	1.13	0.81	1.17	1.48	1.86	1.56
	HDD10.0	1490	212	203	222	203	153	88	23	0	6	59	136	184
	HDD18.3	4451	471	437	480	453	411	338	273	218	225	316	386	443
	CDD10.0	82	0	0	0	0	0	0	8	41	31	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.8	8.3	8.8	7.8	7.5	6.5	6.3	5.9	6.7	8.0	8.5	9.2	9.3
Precipitation	PrecAvg	1154	112	97	78	60	63	70	89	103	125	122	108	127
	PrecMax	1406	170	150	112	108	101	135	135	180	221	194	176	215
	PrecMin	896	49	19	36	16	24	28	52	32	75	69	55	70
	PrecStd	111	27	33	22	20	18	22	23	35	31	32	29	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	6.1	5.6	5.6	7.8	10.4	12.7	14.0	13.6	12.4	10.1	7.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	5.8	5.7	5.2	5.2	6.9	9.7	12.0	13.4	13.0	11.2	9.5	6.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	5.4	5.3	5.0	5.0	6.5	9.0	11.6	13.0	12.8	10.6	8.9	6.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	5.2	5.1	4.7	4.7	6.2	8.5	11.1	12.6	12.5	10.1	8.1	6.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	1.9	2.0	1.9	1.6	1.2	1.0	1.1	1.1	1.3	1.6	2.1	2.1
		MCDBR	1.9	2.0	2.0	1.7	1.3	1.2	1.1	1.2	1.5	1.8	1.8	2.3
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.334	0.352	0.378	0.428	0.408	0.394	0.400	0.398	0.394	0.368	0.347	0.338
	taud		2.285	2.299	2.229	2.109	2.244	2.367	2.409	2.415	2.329	2.291	2.220	2.203
	Ebn at Noon		651	754	805	804	845	861	849	830	783	719	614	560
	Edh at Noon		67	87	112	141	130	116	110	104	101	86	70	62
All-Sky Solar Radiation	RadAvg		0.74	1.46	2.61	3.71	4.38	4.27	3.97	3.41	2.82	1.81	0.93	0.58
	RadStd		0.07	0.11	0.18	0.22	0.41	0.35	0.27	0.32	0.17	0.13	0.08	0.06

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page